PetCare Grooming Co.

student–m051

PetCare Grooming Co. is a small-to-medium pet grooming salon that offers services such as bathing, nail trimming, fur styling, and flea treatments for cats and dogs. Clients can call or visit in person to schedule appointments. The business serves dozens of pet owners weekly and has several groomers working in rotating shifts.

Does the Business Currently Use Any IT Solution for Their Recordkeeping?
Currently, PetCare Grooming Co. keeps records manually using paper forms or basic spreadsheets. Appointment times, pet names, owner contacts, and service details are entered by hand, and updates often result in errors or missed appointments. There is no centralized, searchable system in place.

Why Is the Database Required?
As the number of clients grows, manually managing bookings and customer data becomes inefficient and error-prone. A relational database can help:

- Track pet and owner information consistently.
- Manage appointment scheduling and avoid double-bookings.
- Associate services with specific appointments for accurate billing.
- Provide a historical record of services per pet, which helps with recurring treatments.

What Potential Benefits Will the Business Enjoy by Using the Database?

- Improved Efficiency: Staff can quickly search for pet or owner records, upcoming appointments, and service history.
- Better Customer Service: Employees can greet pets by name, remember preferences, and send appointment reminders.
- Fewer Errors: Reduces missed appointments and incorrect billing due to human error.
- Scalability: As the business grows, the system can handle more clients and services without added complexity.
- Data Insights: Reports can be generated to identify the most popular services, peak booking times, and identify the most loyal customers.

Entities/Tables

1. Owners [Info about the pet owners.]
OwnerID (PK)
FirstName
LastName

Phone
Email
Address
City
State

Why: To store essential information about the pet owners, who are the primary clients of the business. This table allows communication, billing, and service history to be linked to the responsible person.

2. Pets [Each pet belongs to an owner.]
PetID (PK)
OwnerID (FK)
Name
Species
Breed
Age
Weight
MedicalNotes

Why: To store details about each pet receiving grooming services. Since owners can have multiple pets, this table supports a one-to-many relationship and enables tracking of pet-specific needs (e.g., allergies or breed-based grooming styles).

3. Groomers[The groomers working at the salon.]
GroomerID (PK)
FirstName
LastName
Phone
Email
Specialty

Why: To manage employee records for each groomer working in the salon. This allows appointments to be assigned to specific groomers and supports service specialization.

4. Services[The grooming services offered]
ServiceID (PK)
ServiceName
Description
Price
DurationMinutes

Why: To define all available grooming services (e.g., bathing, nail trimming). It standardizes what the salon offers and enables pricing, timing, and description tracking.

5. Appointments[Each visit booked for a pet.]
AppointmentID (PK)
PetID (FK)
GroomerID (FK)
AppointmentDate

Status
Notes

Why: To schedule and track each visit a pet makes to the salon. This ties together pets, groomers, and services, and allows for appointment history, notes, and statuses (e.g., completed, canceled).

6. AppointmentServices M:N relationship between Appointments and Services.
AppointmentID (FK)
ServiceID (FK)
Composite PK

Why: To handle the many-to-many relationship between Appointments and Services. A single appointment can include multiple services, and each service can appear in many appointments. This table supports flexible booking options.

7. Payments[Payment info tied to appointments.]
PaymentID (PK)
AppointmentID (FK)
AmountPaid
PaymentDate
PaymentMethod

Why: To record payment details for appointments. This ensures financial tracking, supports different payment methods, and links payments to specific appointments for auditing and reporting.

8. Products[Retail products sold (e.g., shampoos, brushes).]
ProductID (PK)
ProductName
Description
Price
StockQuantity

Why: To manage the inventory of retail products (e.g., pet shampoo, brushes) sold at the salon. This table tracks pricing, descriptions, and stock levels to support product sales and reordering.

9. Sales[Each product sale to an owner.]
SaleID (PK)
OwnerID (FK)
SaleDate
TotalAmount

Why: To capture transaction-level information when an owner purchases one or more products. This table links each sale to an owner and provides a record of sale date and total amount.

10. SaleDetails [Associative table: products per sale (M:N)]
SaleID (FK)
ProductID (FK)
Quantity
Composite PK

Why: To handle the many-to-many relationship between Sales and Products. A sale can contain multiple products, and a product can appear in multiple sales. This table allows itemized tracking within each transaction.

11. GroomerServices [Associative table: services per groomer (M:N)]
GroomerID (FK)
ServiceID (FK)
Composite PK

Why: To resolve the many-to-many relationship between Groomers and Services. This table indicates which groomers are qualified to perform which services, supporting staff scheduling and filtering services by groomer capabilities.

Entity Relationships and Cardinalities
1. Owners ↔ Pets
Relationship: One-to-Many (1:N)
Explanation: One owner can have many pets, but each pet belongs to one owner.
Min Cardinality:
Owner: 1 (must have at least one pet to use services)
Pet: 1 (must be linked to an owner)

2. Pets ↔ Appointments
Relationship: One-to-Many (1:N)
Explanation: Each pet can have many appointments, each appointment is for one pet.
Min Cardinality:
Pet: 0 (not all pets may have appointments yet)
Appointment: 1 (must be linked to a pet)

3. Groomers ↔ Appointments
Relationship: One-to-Many (1:N)
Explanation: One groomer handles many appointments, each appointment is assigned to one groomer.
Min Cardinality:
Groomer: 0 (a groomer may have no appointments yet)
Appointment: 1 (must have a groomer assigned)

4. Appointments ↔ Services
Relationship: Many-to-Many (M:N)
Why?: Each appointment can include multiple services, and each service can be part of many appointments.
Associative Table: AppointmentServices
Min Cardinality:
Appointment: 1 (must have at least one service)
Service: 0 (some services may not yet be booked)

5. Appointments ↔ Payments
Relationship: One-to-One (1:1)
Explanation: One payment per appointment.

Min Cardinality:
Appointment: 1 (must be paid)
Payment: 1 (must refer to a valid appointment)

6. Products ↔ SaleDetails
Relationship: One-to-Many (1:N)
Explanation: One product can appear in many sale details.
Min Cardinality:
Product: 0 (some products might not be sold yet)
SaleDetail: 1 (must refer to a product)

7. Sales ↔ SaleDetails
Relationship: One-to-Many (1:N)
Explanation: One sale can involve many products.
Min Cardinality:
Sale: 1 (must have at least one line item)
SaleDetail: 1 (must be part of a sale)

8. Owners ↔ Sales
Relationship: One-to-Many (1:N)
Explanation: One owner can make many purchases.
Min Cardinality:
Owner: 0 (may not have bought products)
Sale: 1 (must be tied to a buyer)

9. Groomers ↔ Services
Relationship: Many-to-Many (M:N)
Explanation: Multiple groomers can provide multiple services.
Min Cardinality:
Groomer: 1 (must provide a service)
Service: 1 (must be tied to a groomer)

10. Groomers ↔ GroomerServices
Relationship: One-to-Many (1:N)
Explanation: One groomer can provide multiple services.
Min Cardinality:
Groomer: 1 (must provide a service)
GroomerService: 1 (must belong to a groomer)

11. Services ↔ GroomerServices
Relationship: One-to-Many (1:N)
Explanation: One groomer can provide multiple services.
Min Cardinality:
Service: 1 (must be tied to at least one groomer)
GroomerService: 1 (must reference a valid service)

Associative Tables

1. AppointmentServices

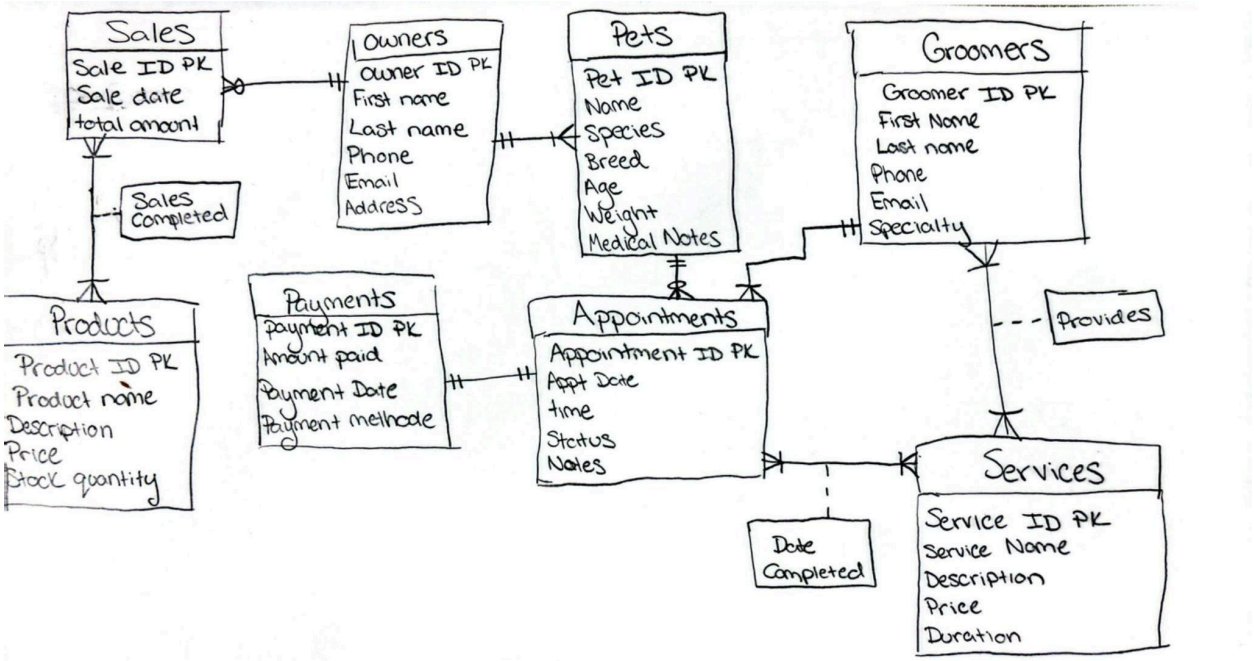
Resolves the many-to-many relationship between Appointments and Services. Each appointment can have multiple services (e.g., bath + nail trim). Each service can appear in multiple appointments.

2. SaleDetails
Resolves the many-to-many relationship between Sales and Products. Each sale can include multiple products. Each product can appear in multiple sales.

3. GroomerServices
Resolves the many-to-many relationship between Groomers and Services. Each groomer can provide multiple services. Each service can appear in multiple groomers.

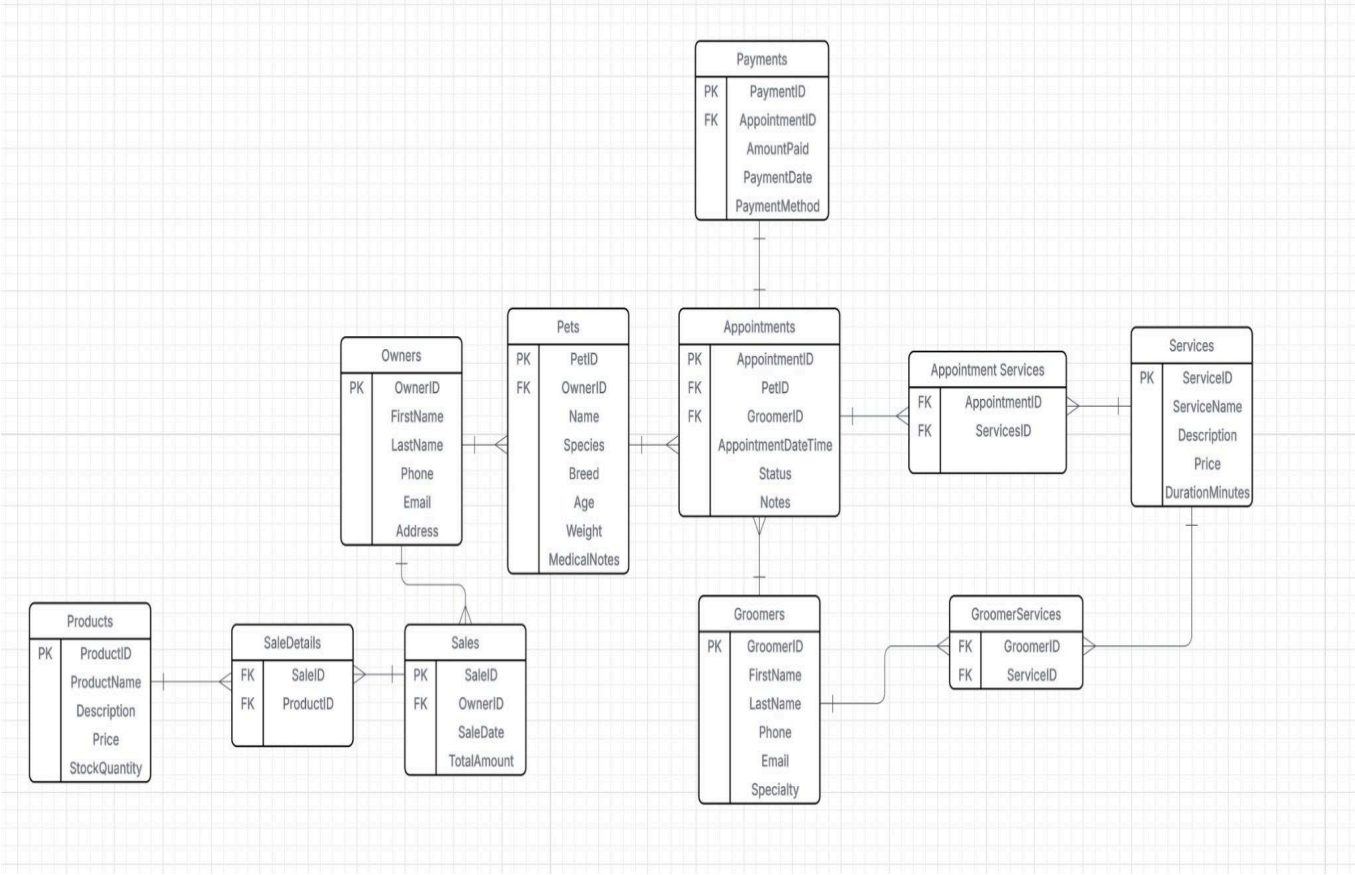
Entity Relationship Diagrams (ERD)

Conceptual Data Model



Scanned with CamScanner

Logical Data Model (LDM)



Queries

Query 1- Total revenue per groomer

Why: The business would want to track individual groomer contributions for performance-based incentives.They can look at this daily, monthly, yearly, etc to see who their top performers are.

```sql
/*Query 1- Total revenue per groomer*/
SELECT g.FirstName, g.LastName, SUM(p.AmountPaid) AS RevenueGenerated
FROM Payments p
INNER JOIN Appointments a
ON p.Appointments_AppointmentID = a.AppointmentID
INNER JOIN Groomers g
ON a.Groomers_GroomerID = g.GroomerID
GROUP BY g.GroomerID;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

FirstName	LastName	RevenueGenerated
Linda	Bates	66
Amy	Smith	49
James	O'Neill	66
Ann	Jacobson	191
Sophie	Pulanski	171
Brittany	Martins	65
Owen	Adams	87
Rachel	Brown	187
Michelle	Jackson	61
Jackie	King	63

Query 2- Pets with Special Medical Notes

Why: Groomers can prepare for animals with allergies or conditions. Groomers need to be aware of any pets that require special medical attention or have conditions. Knowing this ahead of time allows the team to use appropriate products (e.g., hypoallergenic shampoo), schedule extra time if needed, avoid injury or medical emergencies, and improve customer satisfaction by showing care and attentiveness. This query can be used everyday business day or right before any appointment.

```sql
/*Query 2- Pets with Special medical notes*/
SELECT Name, Species, MedicalNotes
FROM Pets
WHERE MedicalNotes IS NOT NULL;
```

esult Grid | Filter Rows: | Export: | W

Name	Species	MedicalNotes
Lucy	Cat	
Rocky	Dog	
Luna	Cat	
Molly	Dog	
Daisy	Dog	
Buddy	Dog	Lyme Disease
Max	Dog	Ringworm
Bella	Cat	Hypothyroidism
Charlie	Dog	
Bailey	Cat	
Oscar	Dog	

Query 3- Most Recent Appointment per Owner and Pet

Why: Identify how long it's been since a pet's last visit and will be helpful for sending reminders to owners who haven't scheduled a visit in a while. This query can be used monthly to see if customers haven't been to the shop in over 30 days.

```sql
/*Query 3- Most Recent Appointment per Owner and Pet*/
SELECT o.FirstName, o.LastName, p.Name AS PetName, MAX(a.AppointmentDate) AS LastAppointment
FROM Owners o
INNER JOIN Pets p
ON o.OwnerID = p.Owners_OwnerID
INNER JOIN Appointments a
ON p.Appointments_AppointmentID = a.AppointmentID
GROUP BY p.PetID;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

FirstName	LastName	PetName	LastAppointment
Alice	Smith	Molly	2025-04-14
Bob	Johnson	Max	2025-04-16
Carol	Williams	Buddy	2025-05-04
David	Brown	Luna	2025-05-02
Eva	Jones	Bella	2025-04-28
Frank	Miller	Rocky	2025-04-22
Grace	Davis	Oscar	2025-03-31
Hannah	Garcia	Lucy	2025-04-25
Ian	Martinez	Bailey	2025-05-01
Jane	Lee	Daisy	2025-04-27
Jane	Lee	Charlie	2025-04-29

Query 4- Owners who have made payments above $50

Why: This query identifies high-spending customers who have made individual payments over $50. These clients are valuable to the business and could be used for targeted loyalty rewards or thank-you messages. It helps segment customers based on spending behavior to improve customer retention, marketing efficiency, and revenue growth. The business can use it monthly for marketing reviews, or quarterly for analyzing customer lifetime value and targeting upsells.

```sql
/*Query 4- Owners who have made payments above $50*/
SELECT DISTINCT o.FirstName, o.LastName, pay.AmountPaid
FROM Payments pay
INNER JOIN Appointments a
ON pay.Appointments_AppointmentID = a.AppointmentID
INNER JOIN Pets p
ON a.AppointmentID = p.Appointments_AppointmentID
INNER JOIN Owners o
ON p.Owners_OwnerID = o.OwnerID
WHERE pay.AmountPaid > 50;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

FirstName	LastName	AmountPaid
Hannah	Garcia	66
David	Brown	66
Alice	Smith	94
Jane	Lee	59
Carol	Williams	65
Bob	Johnson	51
Eva	Jones	92
Jane	Lee	61
Ian	Martinez	63

Query 5- Pets that had an appointment but no payment

Why: This query identifies pets that had a grooming appointment but no payment recorded. This is crucial for detecting missed payments due to errors in billing, it also allows for following up with customers with outstanding balances, and spotting no-show or canceled appointments where service may not have been completed but still logged. It helps prevent revenue loss, improve operational accuracy, and ensure that every service provided is accounted for financially. This should be used daily, to make sure the business didn't miss any transactions.

```sql
/*Query 5- Pets that had an appointment but no payment*/
SELECT Name
FROM Pets
WHERE Appointments_AppointmentID IN
(SELECT AppointmentID
FROM Appointments
WHERE AppointmentID NOT IN (
SELECT Appointments_AppointmentID
FROM Payments));

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Name